

CSC1109 – Data (Analytics) at Speed and Scale

Alessandra Mileo

alessandra.mileo@dcu.ie

Recap: things to remember

- Difference between PIG and HIVE?
- Simple data types examples?
- Complex data types?
- What is Lazy Evaluation in PIG?

Recap: things to remember

- Data Factory
- Data types:
 - Fields are typed values
 - Tuples: ordered set of fields (`field1, field2, ...`)
 - Maps: set of [`key#value`] pairs
 - Bags: collection of tuples `{tuple1, tuple2, ...}`
`{ (a1, a2, ...), (b2, b2, ...), ... }`
- **Lazy evaluation**: nothing happens till a `DUMP` or `STORE`
- Can run locally or on distributed cluster

Note on FLATTEN tuples and bags

- Flatten un-nest tuples, bags and maps:
 - FLATTEN applied to tuples replace the tuples with its fields
 - FLATTEN applied to bags, generate tuples (normally)
 - FLATTEN applied to maps, generates a tuple per Key-Value Pair in the map.
- See guide on FLATTEN operator at <https://pig.apache.org/docs/latest/basic.html>

Note on Schema

- Defined with `LOAD` and `FOREACH` operators using `AS` clause
- Don't have to define `name : type` but you should do so if you want to refer to field by name, otherwise
 - schema is null and fields are treated as bytearray (cast on the fly)
 - you need to refer by positional notation (`$0`, `$1`, etc.)
- Unknown (or null) schema propagates (e.g. `UNION`)
- Complex types can often be automatically inferred correctly:

```
A = LOAD 'data' AS (B: bag {T: tuple(t1:int, t2:int, t3:int)});
```

```
A = LOAD 'data' AS (B: {T: (t1:int, t2:int, t3:int)});
```

Wordcount, again

alias

Field name
with no type

```
lines = LOAD 'local_file.txt' AS (line:chararray);  
words = FOREACH lines GENERATE FLATTEN(TOKENIZE(line)) AS word;  
grouped = GROUP words BY word;  
wordcount = FOREACH grouped GENERATE group, COUNT(words);  
DUMP wordcount;
```

Schema with
no types

Example: UNION and schema

```
a = LOAD 'A.txt' using PigStorage(',') ;  
B = LOAD 'B.txt' using PigStorage(',') as (b1:int, b2:int, b3:int) ;  
C = UNION a,B;
```

```
DUMP C;
```

```
(0,1)  
(1,3)  
(0,5,2)  
(1,7,8)
```

```
DUMP a;  
(0,1)  
(1,3)
```

```
DUMP B;  
(0,5,2)  
(1,7,8)
```

```
Describe C;
```

```
Schema for C is unknown
```

- First schema is null, it propagates through the UNION

Example: UNION and schema

```
B = LOAD 'B.txt' using PigStorage(',') as (b1:int, b2:int, b3:int);  
C = FOREACH B GENERATE b1, b2*b3;  
D = UNION C,B;  
Describe D;  
Schema for D is unknown.
```

- Merging incompatible schema returns null schema

EXAMPLE: UNION and schema

```
A = LOAD 'A.txt' using PigStorage(',') as (a1:int, a2:int, a3:int);
B = LOAD 'B.txt' using PigStorage(',') as (b1:int, b2:int, b3:int);
C = UNION B,A;
D = UNION A,B;
Describe C;
C: {b1:int, b2:int, b3:int}
Describe D;
D: {a1:int, a2:int, a3:int}
```

- Merging compatible schemas takes the first schema.

UNION ON SCHEMA

Combines different schema following general principles:

- Columns with same name appear once in the UNION if compatible:
- Result of type casting is an “escalate” type where priority is
 - double > float > long > int > bytearray
 - tuple|bag|map|chararray > bytearray
- Columns with different names and/or incompatible types are both kept in the resulting schema
- High chance of empty values (for columns present in one schema only)

EXAMPLE: UNION ON SCHEMA

```
A = LOAD 'A.txt' using PigStorage(',') as (a1:int, a2:int, a3:int);
```

```
B = LOAD 'B.txt' using PigStorage(',') as (b1:int, b2:int, b3:int);
```

```
DUMP A;  
(0,1,2)  
(1,3,4)
```

```
DUMP B;  
(0,5,2)  
(1,7,8)
```

```
C = UNION ONSCHEMA A,B;
```

```
DUMP C;
```

EXAMPLE: UNION ON SCHEMA

```
A = LOAD 'A.txt' using PigStorage(',') as (a1:int, a2:int, a3:int);
```

```
B = LOAD 'B.txt' using PigStorage(',') as (b1:int, b2:int, b3:int);
```

```
C = UNION ONSCHEMA A,B;
```

```
DESCRIBE C;
```

```
C: {a1:int, a2:int, a3:int, b1:int, b2:int, b3:int}
```

EXAMPLE: UNION ON SCHEMA

```
L1 = LOAD 'f1' USING (a : int, b : float);
```

```
L2 = LOAD 'f1' USING (a : long, c : chararray);
```

```
U = UNION ONSCHEMA L1, L2;
```

```
DESCRIBE U ;
```

```
U : {a:long, b:float, c:chararray}
```

EXAMPLE: UNION ON SCHEMA

```
L1 = LOAD 'f1' USING (a : int, b : float);
```

```
L2 = LOAD 'f1' USING (a : long, c : chararray);
```

```
U = UNION ONSCHEMA L1, L2;
```

```
DESCRIBE U ;
```

```
U : {a:long, b:float, c:chararray}
```

```
DUMP L1;  
(11,12.0)  
(21,22.0)
```

```
DUMP L2;  
(11,a)  
(12,b)  
(13,c)
```

```
DUMP U;  
(11,12.0,)  
(21,22.0,)  
(11,,a)  
(12,,b)  
(13,,c)
```

Pig Links

- Lab on PIG
 - <https://csc1109.readthedocs.io/en/latest/lab3/>
- Apache Pig Manual: getting started
 - <https://pig.apache.org/docs/r0.11.1/start.html#run>
- PigLatin basics:
 - <https://pig.apache.org/docs/latest/basic.html>